

UCI Shopping EDA

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Link to dataset: <https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset>

```
# Loading data
work_dir <- here()
raw_data <- read_csv(file.path(work_dir, "shopping", "data", "raw_data", "uci_shopping.csv"),
                      show_col_types = FALSE)
head(raw_data)
```

```
## # A tibble: 6 x 18
##   Administrative Administrative_Duration Informational Informational_Duration
##   <dbl>                <dbl>                <dbl>                <dbl>
## 1         0                0                0                0
## 2         0                0                0                0
## 3         0                0                0                0
## 4         0                0                0                0
## 5         0                0                0                0
## 6         0                0                0                0
## # i 14 more variables: ProductRelated <dbl>, ProductRelated_Duration <dbl>,
## #   BounceRates <dbl>, ExitRates <dbl>, PageValues <dbl>, SpecialDay <dbl>,
## #   Month <chr>, OperatingSystems <dbl>, Browser <dbl>, Region <dbl>,
## #   TrafficType <dbl>, VisitorType <chr>, Weekend <lgl>, Revenue <lgl>
```

Description of Variables

Continuous

- **Administrative:** Number of pages visited about account management
- **Administrative Duration:** Total amount of time (in seconds) spent on account management related pages
- **Informational:** Number of pages visited about website, communication and address information of the shopping site
- **Informational Duration:** Total amount of time spent on informational pages
- **Product Related:** Number of pages visited about product related pages
- **Product Related Duration:** Total amount of time spent on product related pages
- **Bounce Rate:** Average bounce rate value of the pages visited

- Formula: $BounceRate = 1 - EngagementRate$
- $EngagementRate$ = percentage of sessions that either lasted longer than 10 seconds, had key events, or two or more screen or page views
- Google Analytics assigns this value to every page of a website
- This feature takes the average of all the bounce rates value from all the pages the user visits in a single session
- **Exit Rate:** Average exit rate value of the pages visited
 - Formula: $ExitRate = \frac{NumberOfExits}{NumberOfPageViews} \times 100$
- **Page Value:** Average page value of the pages visited
 - Formula: $PageValue = \frac{TotalRevenue + TotalGoalValue}{UniquePageViews}$
 - $TotalRevenue$ = amount of money generated from a website page (i.e. transaction page)
 - $TotalGoalValue$ = value assigned to a specific page that is defined by the business
 - $UniquePageViews$ = number of unique user visits, only counted once per session
- **Special Day:** Closeness of the site visiting time to a special day

Categorical

- **OperatingSystems:** Operating system of the visitor
- **Browser:** Browser of the visitor
- **Region:** Geographic region from which the session has been started by the visitor
- **TrafficType:** Traffic source by which the visitor has arrived at the website
- **VisitorType:** Visitor type as “New Visitor”, “Returning Visitor”, and “Other”
- **Weekend:** Boolean value indicating whether the date of the visit is a weekend
- **Month:** Month value of the visit date
- **Revenue:** Class label indicating whether the visit has been finalized with a transaction

Summary

```
continuous_var <- c("Administrative", "Administrative_Duration", "Informational",
                    "Informational_Duration", "ProductRelated", "ProductRelated_Duration",
                    "BounceRates", "ExitRates", "PageValues")

categorical_var <- c("Month", "OperatingSystems", "Browser", "Region", "TrafficType",
                    "VisitorType", "Weekend", "Revenue")

shopping_uci <- raw_data %>%
  mutate(across(all_of(categorical_var), as.factor))

summary(shopping_uci)
```

```
## Administrative Administrative_Duration Informational
## Min. : 0.000 Min. : 0.00 Min. : 0.0000
## 1st Qu.: 0.000 1st Qu.: 0.00 1st Qu.: 0.0000
## Median : 1.000 Median : 7.50 Median : 0.0000
## Mean : 2.315 Mean : 80.82 Mean : 0.5036
```

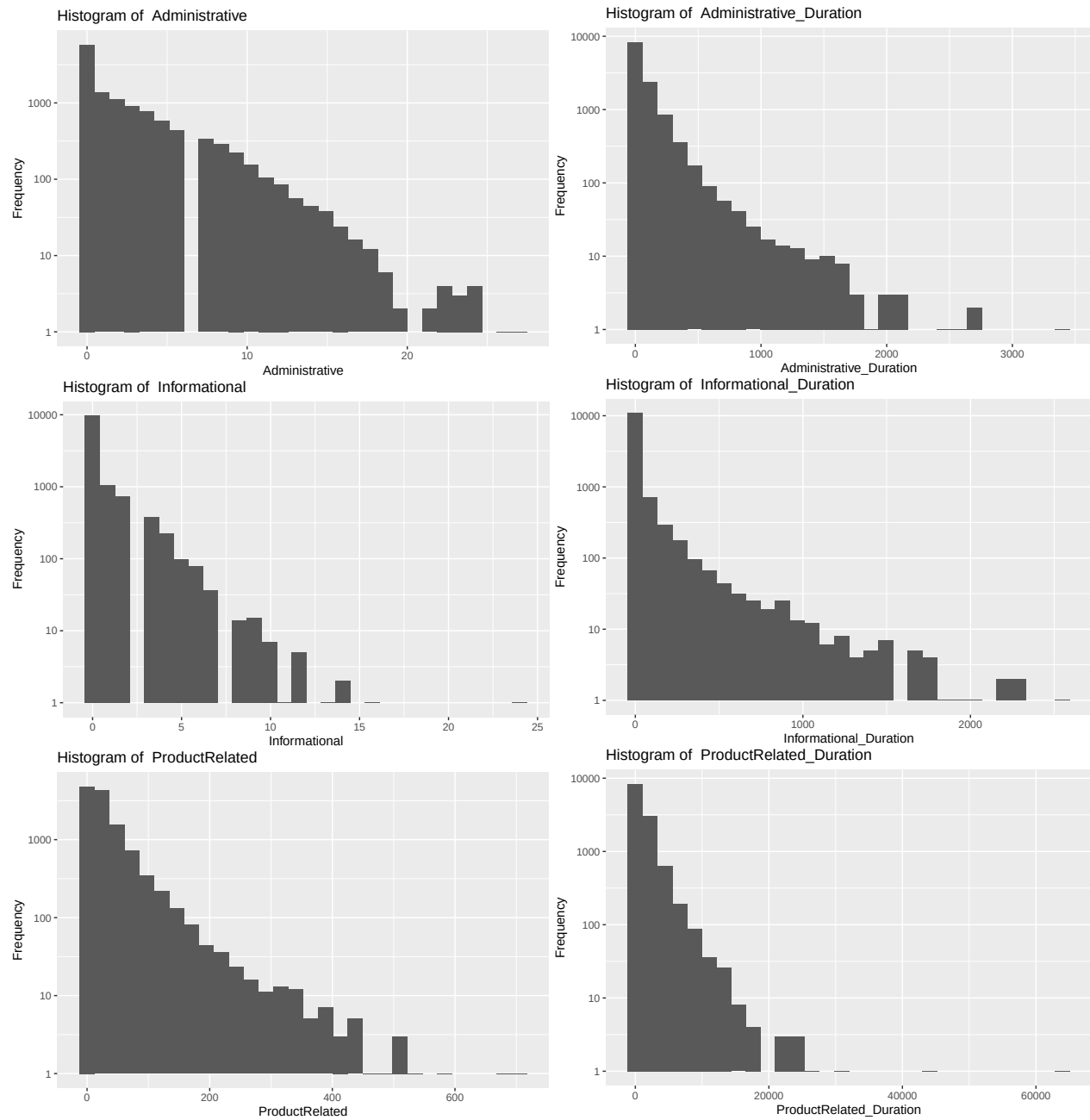
```
## 3rd Qu.: 4.000    3rd Qu.: 93.26        3rd Qu.: 0.0000
## Max.    :27.000    Max.    :3398.75      Max.    :24.0000
##
## Informational_Duration ProductRelated    ProductRelated_Duration
## Min.    : 0.00        Min.    : 0.00    Min.    : 0.0
## 1st Qu.: 0.00        1st Qu.: 7.00    1st Qu.: 184.1
## Median : 0.00        Median : 18.00   Median : 598.9
## Mean    : 34.47       Mean    : 31.73   Mean    : 1194.8
## 3rd Qu.: 0.00        3rd Qu.: 38.00   3rd Qu.: 1464.2
## Max.    :2549.38      Max.    :705.00   Max.    :63973.5
##
## BounceRates      ExitRates      PageValues      SpecialDay
## Min.    :0.000000  Min.    :0.00000  Min.    : 0.000  Min.    :0.00000
## 1st Qu.:0.000000  1st Qu.:0.01429  1st Qu.: 0.000  1st Qu.:0.00000
## Median :0.003112  Median :0.02516  Median : 0.000  Median :0.00000
## Mean    :0.022191  Mean    :0.04307  Mean    : 5.889  Mean    :0.06143
## 3rd Qu.:0.016813  3rd Qu.:0.05000  3rd Qu.: 0.000  3rd Qu.:0.00000
## Max.    :0.200000  Max.    :0.20000  Max.    :361.764 Max.    :1.00000
##
##      Month      OperatingSystems      Browser      Region      TrafficType
## May      :3364    2      :6601    2      :7961    1      :4780    2      :3913
## Nov      :2998    1      :2585    1      :2462    3      :2403    1      :2451
## Mar      :1907    3      :2555    4      : 736    4      :1182    3      :2052
## Dec      :1727    4      : 478    5      : 467    2      :1136    4      :1069
## Oct      : 549    8      : 79     6      : 174    6      : 805   13      : 738
## Sep      : 448    6      : 19    10      : 163    7      : 761   10      : 450
## (Other):1337    (Other): 13    (Other): 367  (Other):1263  (Other):1657
##      VisitorType      Weekend      Revenue
## New_Visitor      : 1694    FALSE:9462    FALSE:10422
## Other              : 85     TRUE :2868     TRUE : 1908
## Returning_Visitor:10551
##
##
##
##
```

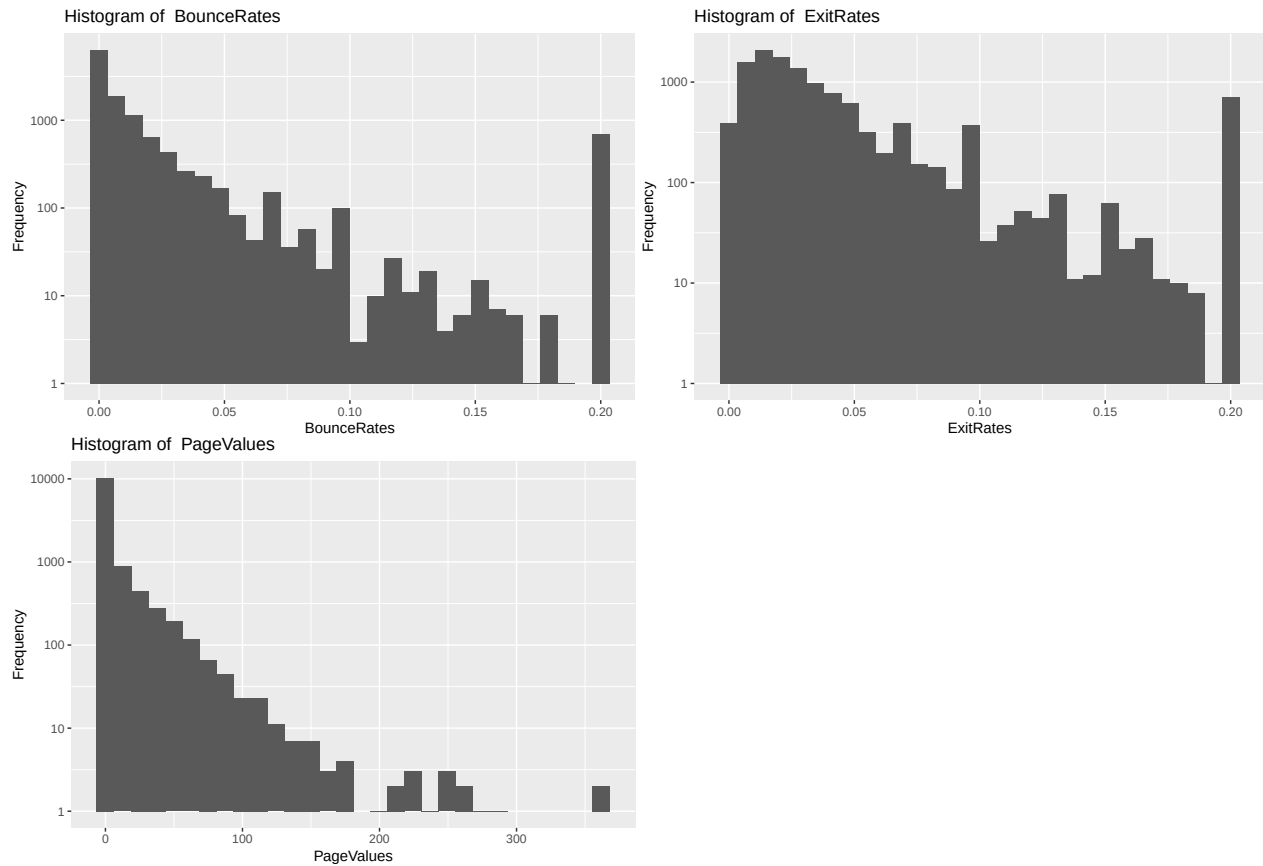
Bar Charts/Histograms of Variables

```
# Histograms of Continuous Variables
par(mfrow = c(2, 3))

for (i in 1:length(continuous_var)) {
  current = shopping_uci[[continuous_var[i]]]
  plot <- ggplot(data = shopping_uci, mapping = aes(x = current)) +
    geom_histogram() +
    labs(
      y = "Frequency",
      x = continuous_var[i],
      title = paste("Histogram of ", continuous_var[i])
    ) +
    scale_y_log10()
}
```

```
print(plot)
}
```





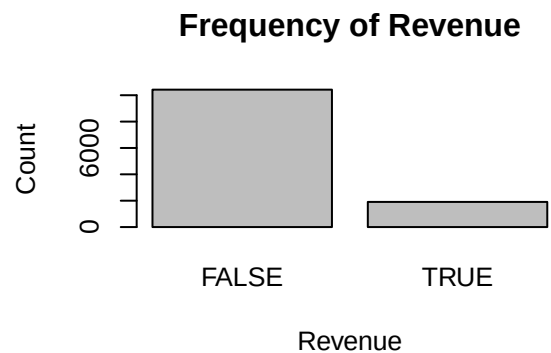
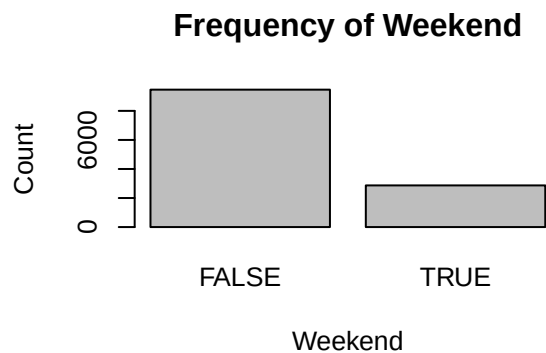
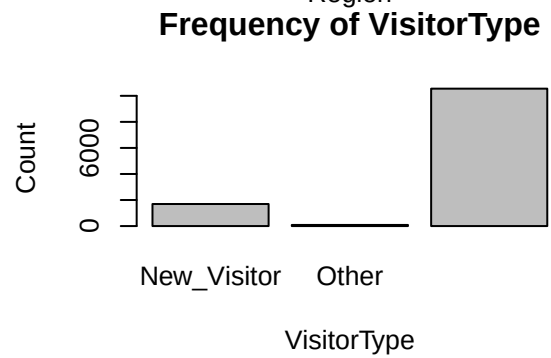
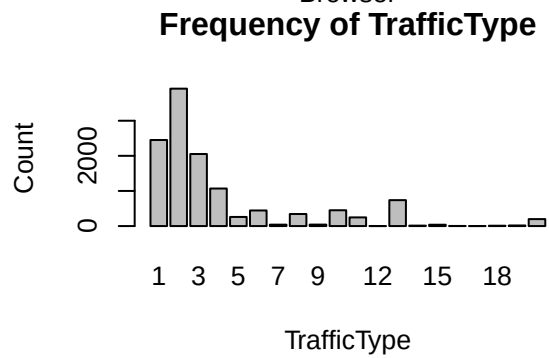
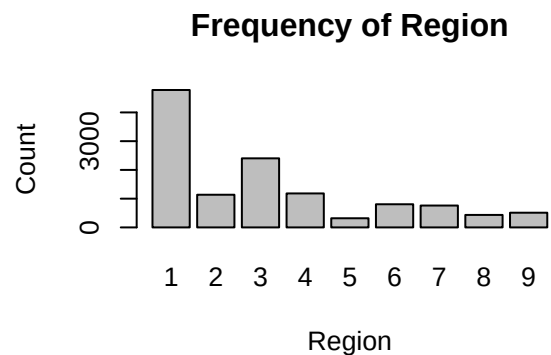
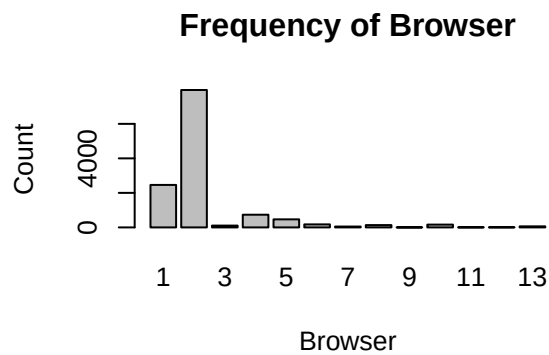
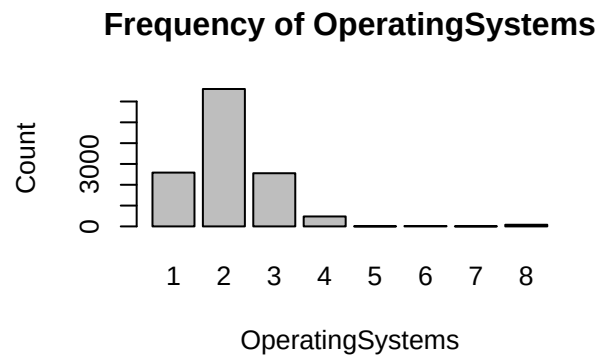
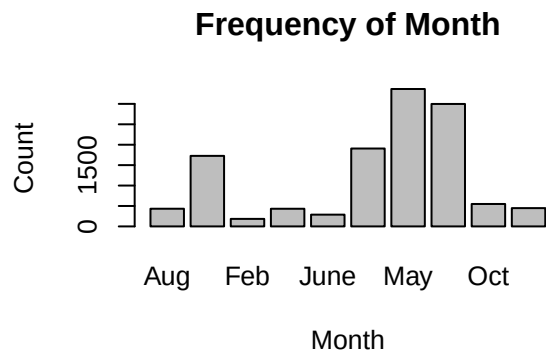
Notes:

- I logged the scale of the y-axes because they were all heavily zero-inflated
- After log transforming the scale, it appears the duration graphs follow a somewhat exponential distribution
- The rates appear to be a little left skewed

```
# Bar Charts of Categorical Variables
par(mfrow = c(2, 2))

for (i in 1:length(categorical_var)) {
  current_var_count <- shopping_uci %>%
    group_by(across(all_of(categorical_var[i]))) %>%
    summarize(count = n())

  barplot(height = current_var_count$count,
          names.arg = current_var_count %>% pull(1),
          xlab = categorical_var[i],
          ylab = "Count",
          main = paste("Frequency of", categorical_var[i]))
}
```



Seeing if combining durations is useful

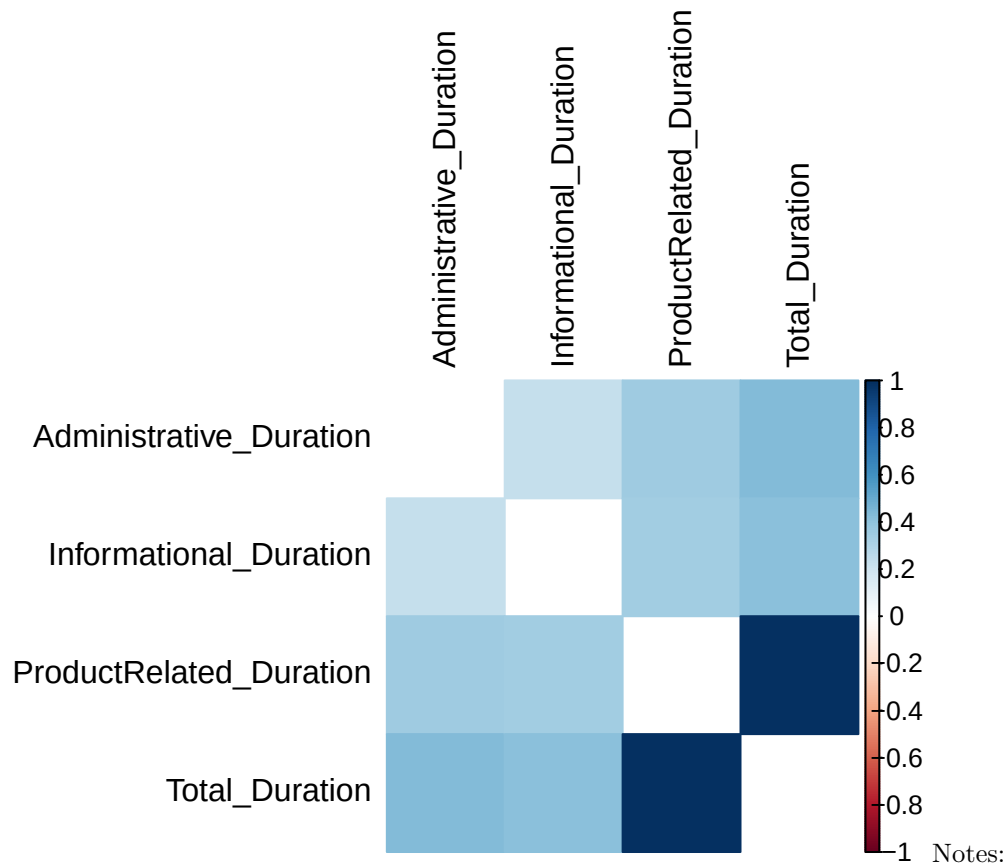
```
# Creating new dataset with total durations
shopping_uci_total_durations <- shopping_uci %>%
  mutate(Total_Duration = Administrative_Duration + Informational_Duration + ProductRelated_Duration)

shopping_uci_total_durations

## # A tibble: 12,330 x 19
##   Administrative_Duration Informational_Duration ProductRelated_Duration Total_Duration
##   <dbl> <dbl> <dbl> <dbl>
## 1      0      0      0      0
## 2      0      0      0      0
## 3      0      0      0      0
## 4      0      0      0      0
## 5      0      0      0      0
## 6      0      0      0      0
## 7      0      0      0      0
## 8      1      0      0      1
## 9      0      0      0      0
## 10     0      0      0      0
## # i 12,320 more rows
## # i 15 more variables: ProductRelated_Duration <dbl>, BounceRates <dbl>, ExitRates <dbl>, PageValues <dbl>, SpecialDay <dbl>,
## #   Month <fct>, OperatingSystems <fct>, Browser <fct>, Region <fct>,
## #   TrafficType <fct>, VisitorType <fct>, Weekend <fct>, Revenue <fct>,
## #   Total_Duration <dbl>

# Checking to see if total duration correlates with other individual durations
corr_values <- shopping_uci_total_durations %>%
  select(c(Administrative_Duration, Informational_Duration, ProductRelated_Duration, Total_Duration)) %>%
  cor()

corrplot(
  corr_values,
  diag = FALSE,
  method = "color",
  tl.col = "black"
)
```

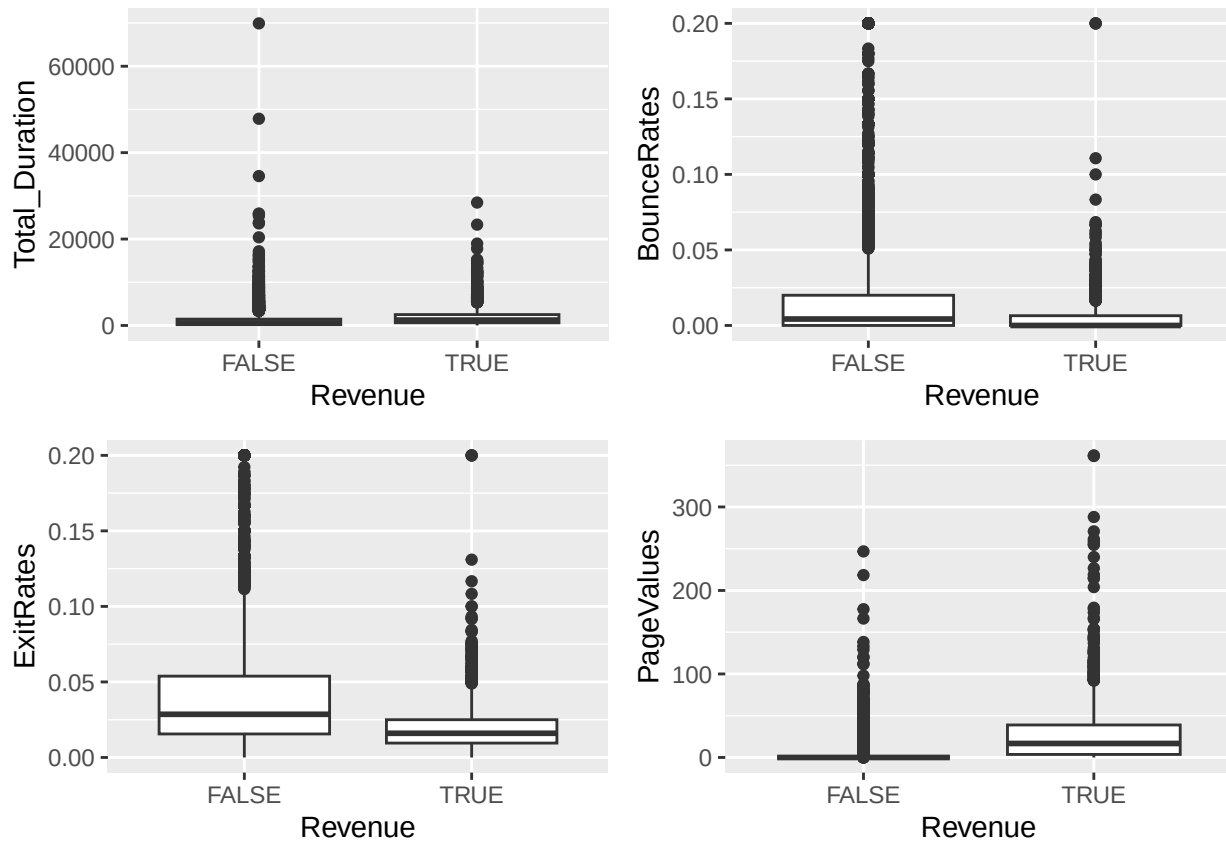


- **Total_Duration** seems to correlate with the other individual durations, especially **ProductRelated_Duration**
- From this, it may be safe to sum up the individual durations into one variable, as it looks like we would not lose that much information if we perform this operation

Multivariate analysis

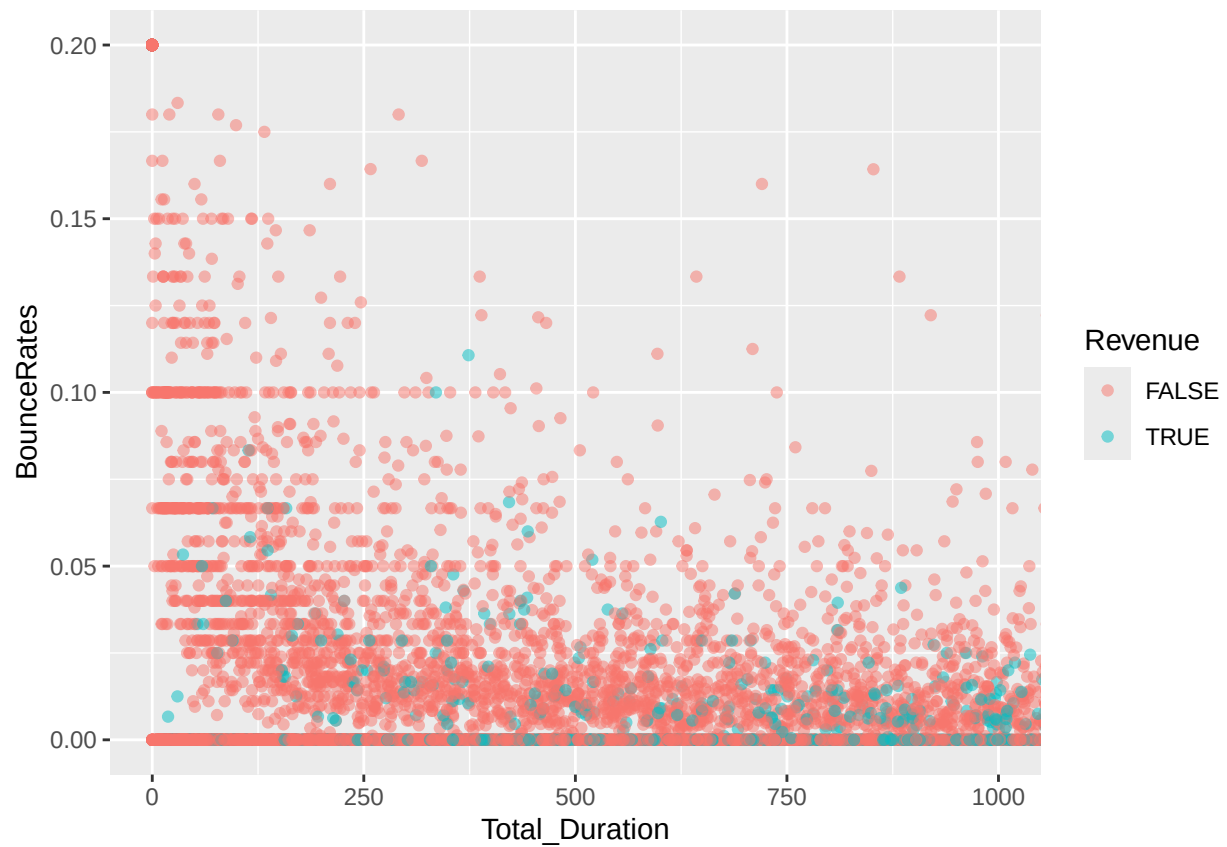
```
# Box plots
total_duration <- ggplot(data = shopping_uci_total_durations, mapping = aes(x = Revenue, y = Total_Duration)) +
  geom_boxplot()
bounce <- ggplot(data = shopping_uci_total_durations, mapping = aes(x = Revenue, y = BounceRates)) +
  geom_boxplot()
exit <- ggplot(data = shopping_uci_total_durations, mapping = aes(x = Revenue, y = ExitRates)) +
  geom_boxplot()
page <- ggplot(data = shopping_uci_total_durations, mapping = aes(x = Revenue, y = PageValues)) +
  geom_boxplot()

grid.arrange(total_duration, bounce, exit, page, ncol = 2)
```

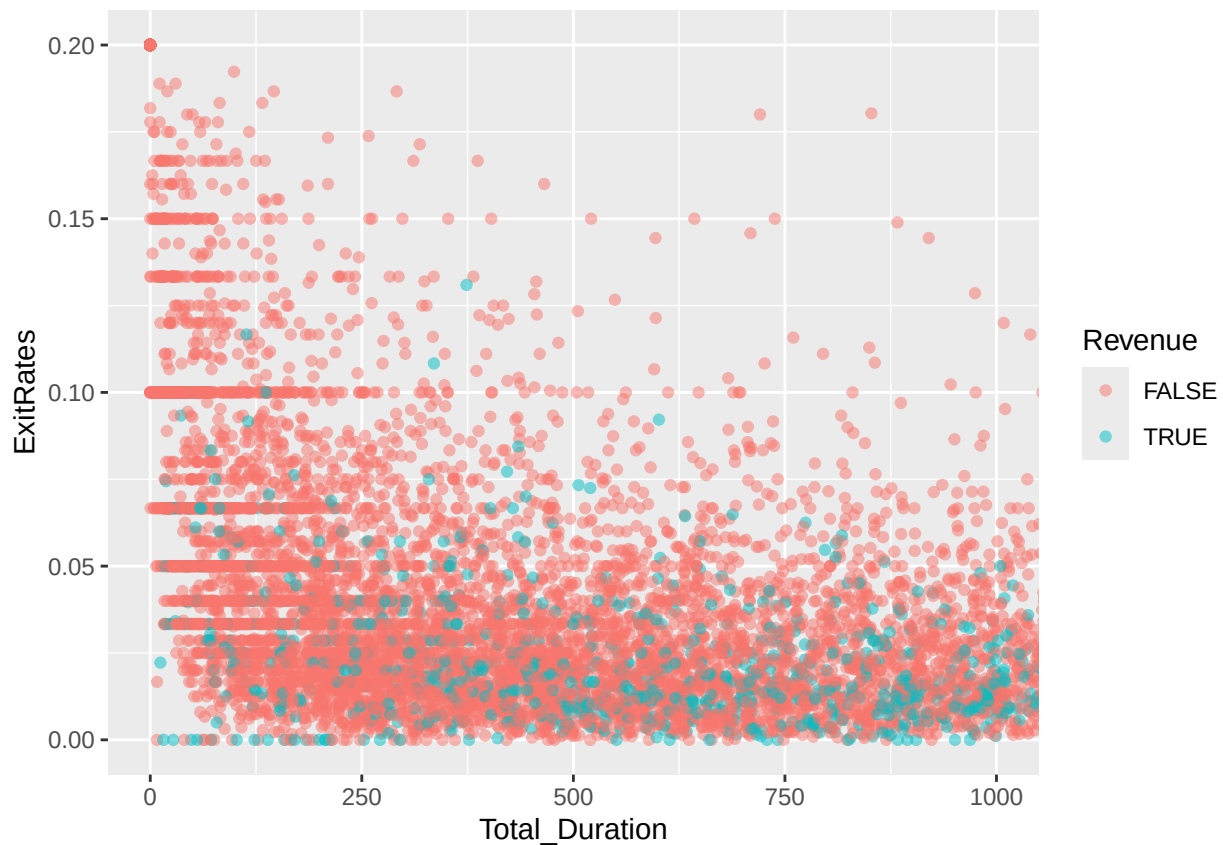



Nothing really useful here.

```
# Scatter plots
ggplot(data = shopping_uci_total_durations, mapping = aes(x = Total_Duration, y = BounceRates,
                                                         color = Revenue)) +
  geom_point(alpha = 0.5) +
  coord_cartesian(xlim = c(0, 1000))
```



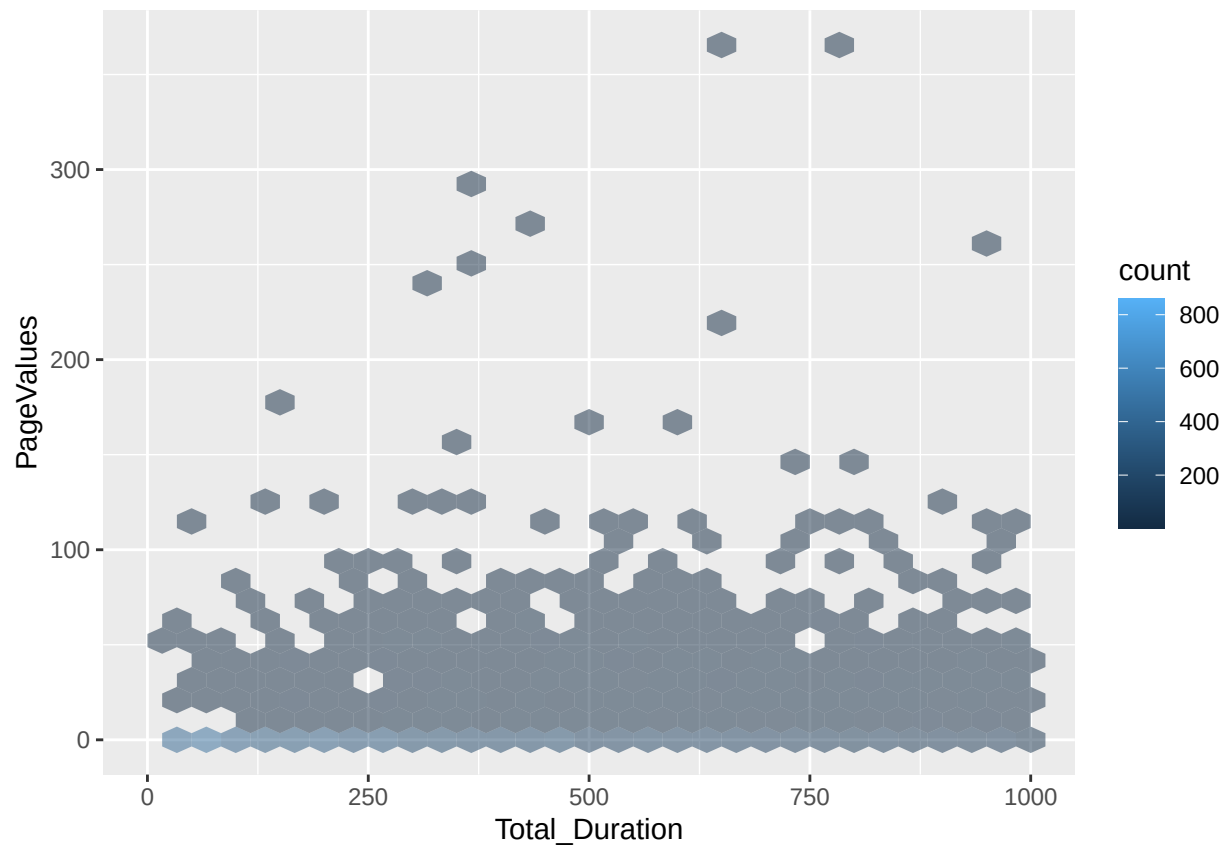
```
ggplot(data = shopping_uci_total_durations, mapping = aes(x = Total_Duration, y = ExitRates,  
                                                         color = Revenue)) +  
  geom_point(alpha = 0.5) +  
  coord_cartesian(xlim = c(0, 1000))
```



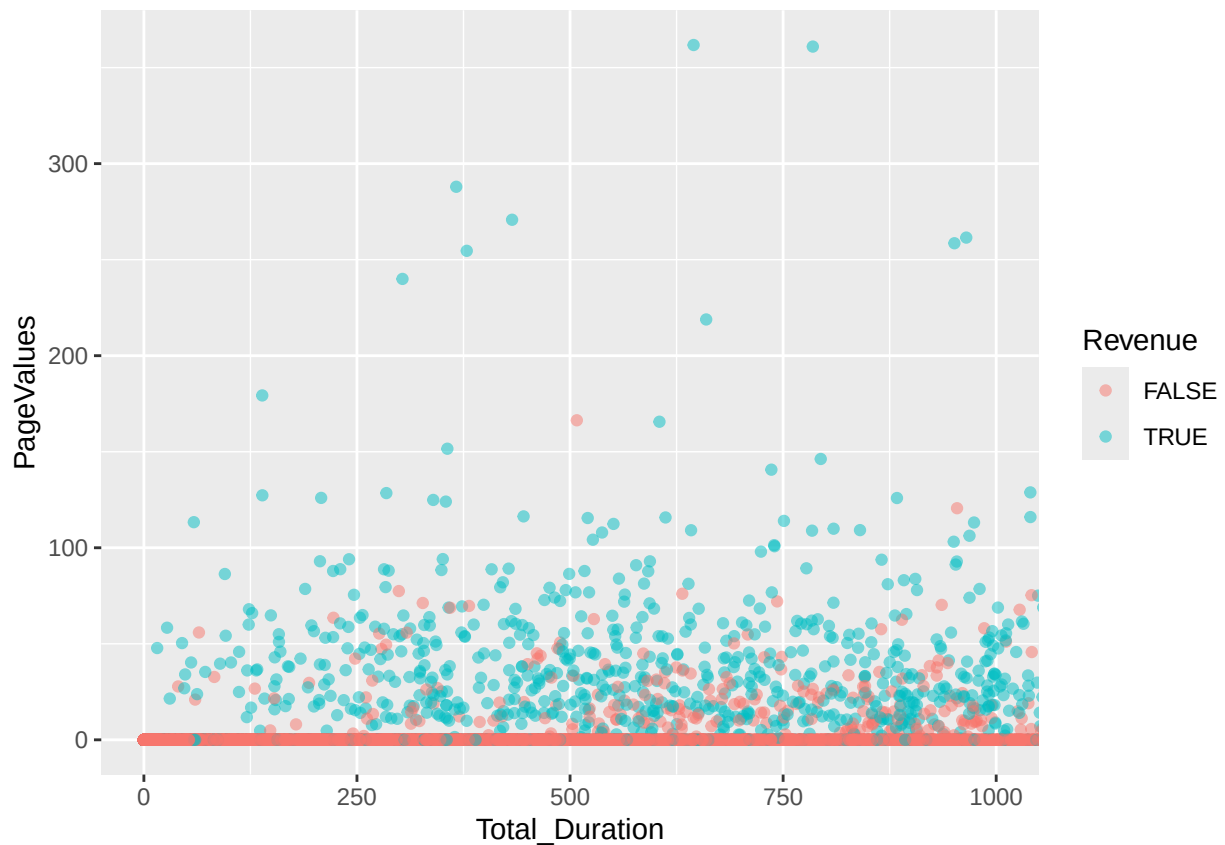
```
shopping_uci_total_durations |>
  ggplot(mapping = aes(x = Total_Duration, y = PageValues ))+
  geom_hex(alpha = 0.5) +
  xlim(0, 1000)
```

```
## Warning: Removed 4768 rows containing non-finite outside the scale range
## ('stat_binhex()').
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_hex()').
```



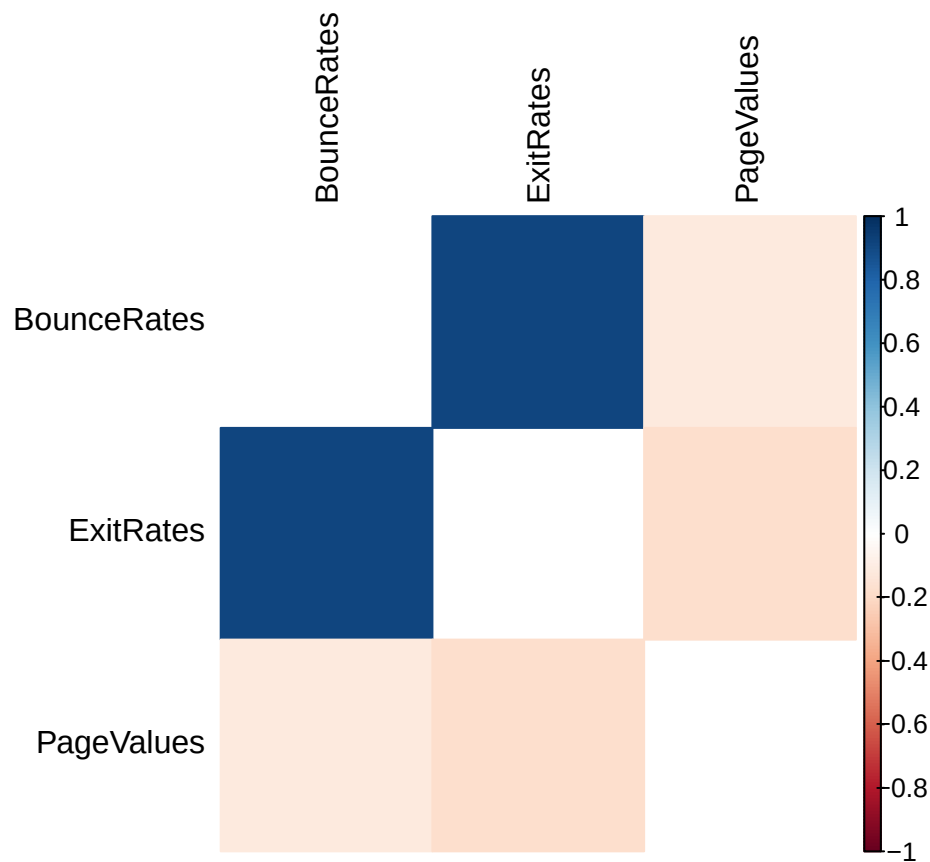
```
ggplot(data = shopping_uci_total_durations, mapping = aes(x = Total_Duration, y = PageValues,  
  color = Revenue)) +  
  geom_point(alpha = 0.5) +  
  coord_cartesian(xlim = c(0, 1000))
```



Nothing really useful can be extracted from this as well

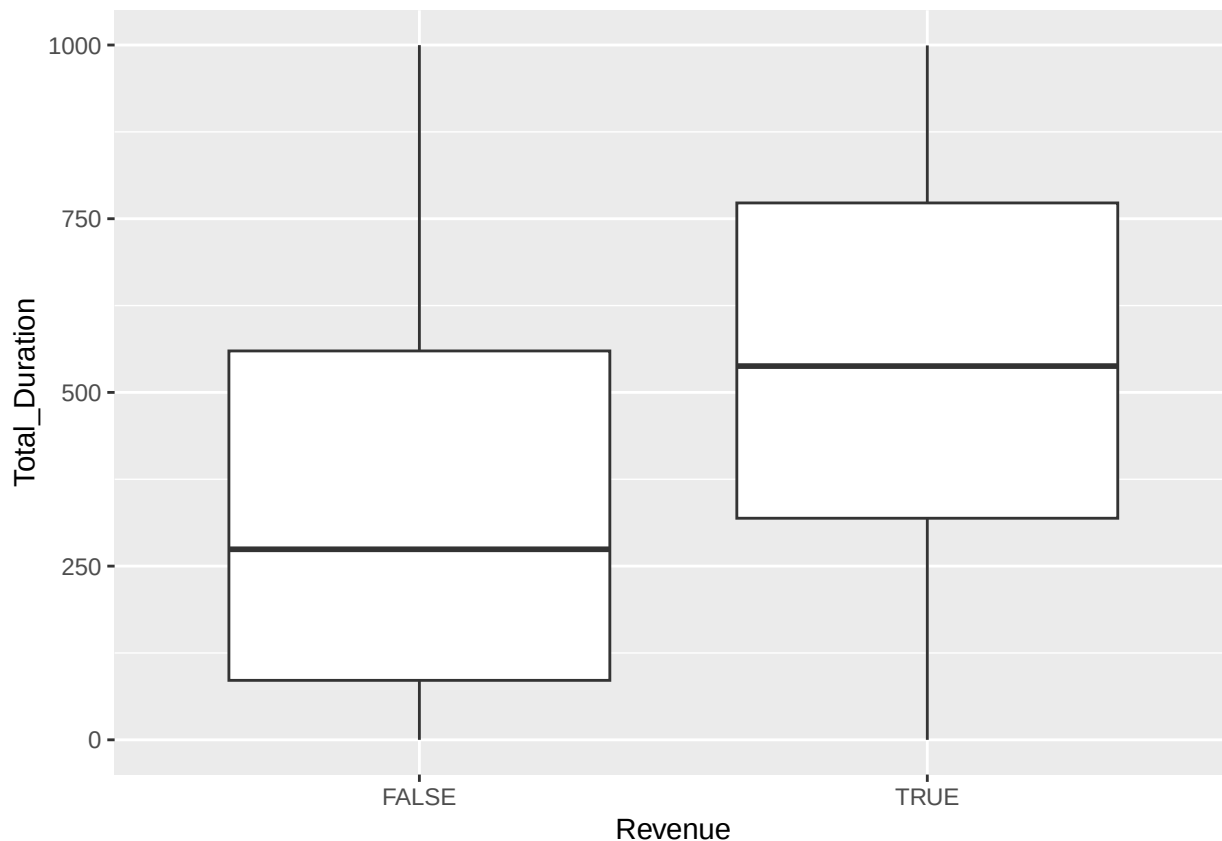
```
# Correlation plot between rates
corr_values <- shopping_uci_total_durations %>%
  select(c(BounceRates, ExitRates, PageValues)) %>%
  cor()

corrplot(
  corr_values,
  diag = FALSE,
  method = "color",
  tl.col = "black"
)
```



```
ggplot(data = shopping_uci_total_durations, mapping = aes(x = Revenue, y = Total_Duration)) +
  geom_boxplot() +
  ylim(0, 1000)
```

```
## Warning: Removed 4768 rows containing non-finite outside the scale range
## ('stat_boxplot()').
```



EDA with cleaned data

TODO: run some graphs on the cleaned data before transforming it

```
# Read preprocessed data
uci_transformed <- read_csv(file.path(work_dir, "shopping", "data", "intermediate_data",
                                     "01_uci_preprocessed.csv"))
```

```
## Rows: 12330 Columns: 19
## -- Column specification -----
## Delimiter: ","
## chr (2): Month, VisitorType
## dbl (15): Administrative, Informational, ProductRelated, BounceRates, ExitRa...
## lgl (2): Weekend, Revenue
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
head(uci_transformed)
```

```
## # A tibble: 6 x 19
```

```
## Administrative Informational ProductRelated BounceRates ExitRates PageValues
## <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
## 1 0 0 1 0.2 0.2 0
## 2 0 0 2 0 0.1 0
## 3 0 0 1 0.2 0.2 0
## 4 0 0 2 0.05 0.14 0
## 5 0 0 10 0.02 0.05 0
## 6 0 0 19 0.0158 0.0246 0
## # i 13 more variables: SpecialDay <dbl>, Month <chr>, OperatingSystems <dbl>,
## # Browser <dbl>, Region <dbl>, TrafficType <dbl>, VisitorType <chr>,
## # Weekend <lgl>, Revenue <lgl>, TotalDuration <dbl>, TimeThreshold <dbl>,
## # CensorTime <dbl>, Censor <dbl>
```

Description of variables

- **TotalDuration**: sum of the **Administrative_Duration**, **Informational_Duration**, and **ProductRelated_Duration**
- **TimeThreshold**: time cutoff; obtained by randomly sampling uniformly from [10, 1800]
- **CensorTime**: total duration of time with censoring; obtained by getting minimum of **TotalDuration** and **TimeThreshold**
- **Censor**: indicator variable signaling censorship

EDA